Fiji Meteorological Service

SYSTEMS ANALYSIS AND DESIGN DOCUMENT

GROUP 13

Flood Risk Prediction and Community Alert System -AI Powered for Fiji MET

Project Team	Role / Main Area
Parvish Mohan	Project Manager / AI-ML / Backend
Rishwant Kumar	Backend / Technical Design
Aryan	Scrum Master / Frontend-QA / Documentation
Louvatu	Frontend / QA
Atharv	DevOps / Database

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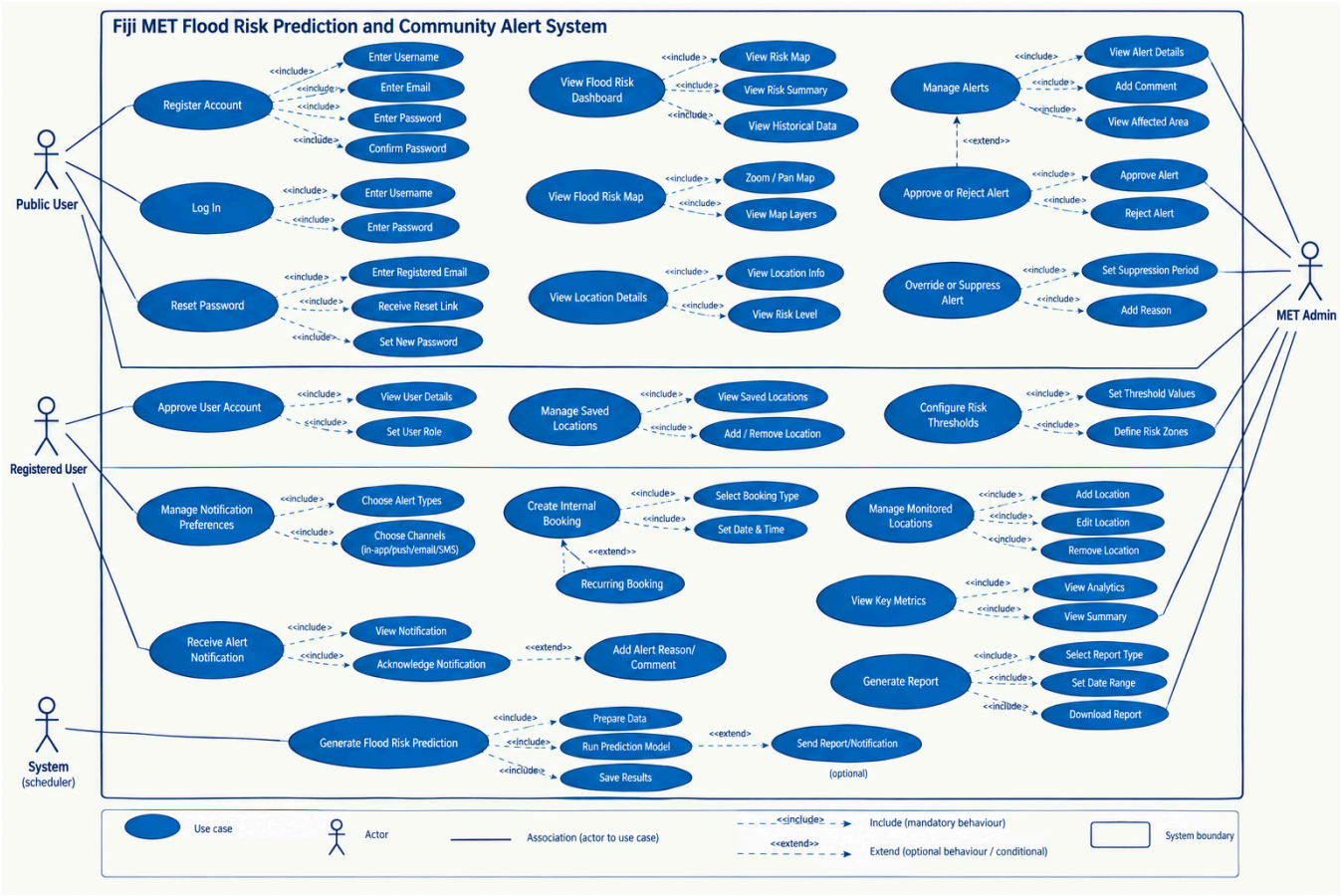
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1. Introduction

The following document presents the system ana;aysis and design for the Flood Risk Prediction and Community Alert System, a web-based application developed for the Fiji Meteorological Services (Fiji MET). The system uses artificial intelligence to predict flood risk from weather and environmental data, and communicates this information to Fiji MET administrators and the public through a dashboard, an interactive flood-risk map, and an alert notification sub-system, while ensuring integrity, scalability and ease of use for all its users.

This document serves as a guide for both the development team and stakeholders in understanding the system's detailed architecture in understanding the system's detailed architecture and its reflection of the system's requirements. Sections 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 would be of interest to the development team while sections 3, 7, 8, 9,12, and 13 would be of interest to stakeholders who lack advanced technical knowledge, in order to understand the scope of the Flood Risk Prediction and Community Alert System.

2. Use Case Diagram



3.0 Use Case Descriptions

3.1 Authentication / Access (if still in approved scope)

Use Case Name	Authentication
Scenario	A Fiji MET administrator will review requests and give access to authenticate users registering for the first time.
Triggering Event	Admin opens the pending requests list.

Brief Description	The admin will see a list of users with pending registration requests. The admin will also have the ability to authenticate users registering for the first time to give access to specific site functions.
Primary Actor	Fiji MET Admin (Granting Access)
Secondary Actor(s)	Users (Requesting Access)
Stakeholders	Fiji MET Admin
Pre-conditions	The user successfully completes the registration form and meets all requirements.
Post-conditions	User Account is granted access and tagged with appropriate authentication level.
Flow of Activities	Actor: 1. Admin navigates to pending request page 2. Admin views a list of pending account requests. 3. Admins clicks on the “accept” button. System: 1.1. System Displays users registration request after user has successfully filled form. 2.1 System displays reject and accept buttons next to user request. 3.1 On the user end, the system displays a “pending” next to account status. 4.1 The system notifies users of a successful registration.
Exception Conditions	1. The account already exists. System displays error. 2. Users

3.2 Flood Risk Dashboard

Use Case Name	Flood Risk Dashboard
Scenario	The system users have access to a dashboard with important information regarding flood risk warnings and risk levels. This dashboard combines and summarises information into an all-in-one page for ease of use.

Triggering Event	Users log in and land on or can navigate to the Flood Risk Dashboard.
Brief Description	The systems contain summarised information regarding flood risk levels, active alerts and other meteorological information. Also included is a live map with live risk level indicators on locations displaying real-time flood risk data.
Primary Actor	Users (Public/ Fiji MET staff)
Secondary Actor(s)	Admins, Fiji MET Staff
Stakeholders	Public Users (at risk communities), Fiji MET staff
Pre-conditions	The user is logged in. The user has at least one saved location, dashboard prompts user to add a location.
Post-conditions	The system displays all required information onto the user flood risk dashboard. Dashboard also displays active alerts and notices based on saved locations and risk levels.=
Flow of Activities	Actor: 1. The User logs in and lands on the Flood Risk Dashboard. 2. The User reviews meteorological summary and any listed active notices/alerts. 3. The User can navigate and interact with the live map. 4. The User can save specific locations and can also view information regarding saved locations. System: 1.1The system receives a list of users' saved locations. 2.1 The system can display current flood-risk predictions based on the user's saved location. 3.1The system can retrieve and display any alerts in relation to the user's location. 2.1 The system can display risk levels on an interactive map. These indicators are colour coordinated in relation to the level of the flood risk level.
Exception Conditions	1. System displays prompt "please add a location" to users who have yet to save a location. 2.System displays risk levels with last known timestamp when a risk data is temporarily unavailable. 3. The system displays a message "unable to load. Please try again" when the database or server is unavailable.

3.3 Weather Data Management

3.3.1 Take In & Store Weather Data

Use Case Name	Take in & Store Weather Data
Scenario	On an Interval, the system fetches weather data for the monitored locations from an External API, validates it and stores the up to date and current weather data.
Triggering Event	A set timer which is automatic built into the system, triggers this fetching of weather data at set intervals.
Brief Description	The backend of the system fetches and receives weather data at periodical intervals from an External API. These weather data are for the active monitored locations and before storing records into PostgreSQL, all the data is validated and recorded with appropriate attributes like timestamps and then stored in the database.
Primary Actor	System-Backend Service(Fast API)
Secondary Actor(s)	External Weather API & PostgreSQL
Stakeholders	Fiji MET Staff & Public Users
Pre-Conditions	1. Selected monitored locations are selected and existing in the database. 2. Weather Data External API's are available and accessible. 3. PostgreSQL database runs and is ready to store weather data.
Post-Conditions	1. Weather data is recorded with correct attributes like timestamps and they are validated before storing in the database. 2. The new set of weather data is made available for the AI Prediction Model.
Flow of Activities	Actor 1.1 The automated timer/scheduler starts the fetching of weather data process. System 1. 1 Gets the list of the Active Monitored Locations currently from the Database. 1.2 From the Weather API, it fetches the weather data for the relevant Active Monitored Locations. 1.3 Gets the data and timestamps it accordingly. 1.4 Validates each data being fetched and ensuring it is within the possible data ranges for attributes like temperature and rainfall. 1.5 Saves all verified weather data in the Database in

	PostgreSQLm table.
Exception Conditions	1. The External API fails to connect: The system acknowledges the disconnection, keeps the last updated weather data in the system and shown on the front-end while it waits for the next time(based upon the interval) to try to reconnect and get current data sets. 2. Invalid Weather Data set Fetched: the system recognizes the invalid data sets and data ranges, disregards that particular data set and continues operation with the last updated data set and waits till the next scheduled update.

3.4 AI Flood Risk Prediction

Use Case Name	Generate Flood Risk Prediction
Scenario	A user or Fiji MET staff member requests the current flood-risk prediction for a monitored location.
Triggering Event	The user selects a monitored location or the application requests the latest prediction for a location.
Brief Description	The system retrieves the latest available weather/location features, prepares them for the trained AI model, runs inference, classifies the flood risk, records the confidence and timestamp, stores the result, and returns it to the requesting interface.
Primary Actor	User / Fiji MET Staff
Secondary Actor(s)	Flood Risk Dashboard; Prediction Service; AI Model; Database
Stakeholders	Fiji MET, public users, at-risk communities, disaster-response agencies, system administrators
Pre-conditions	1. The monitored location exists and is active. 2. Sufficient input data is available for the location. 3. A valid trained prediction model is available to the backend. 4. The backend and database services are operational.
Post-conditions	1. A prediction is produced with a risk category, confidence value and timestamp. 2. The prediction is stored and linked to the monitored location. 3. The result is available to the dashboard/map and may be used by the alert function.

Flow of Activities	ACTOR 1. Selects a monitored location or opens a view that requires a current risk prediction. 2. Requests/refreshes the flood-risk result. 3. Reviews the displayed risk category, confidence and timestamp. SYSTEM 1.1 Validates the location/request. 1.2 Retrieves the latest available weather and location-related input data. 1.3 Cleans/transforms the values into the feature format expected by the model. 2.1 Loads/uses the approved trained model and performs inference. 2.2 Converts the model output into LOW, MODERATE, HIGH or VERY_HIGH. 2.3 Calculates/returns the associated confidence value. 2.4 Stores the prediction, model reference/version where available, and timestamp. 2.5 Returns the prediction to the requesting interface. 3.1 Displays the prediction in a readable form and makes it available to related map/alert functions.
Exception Conditions	1. If required input data is missing or invalid, the system shall not fabricate a prediction and shall return a data-unavailable status. 2. If the model cannot be loaded or inference fails, the system shall log the error and return a safe error response. 3. If the database is temporarily unavailable, the system shall avoid reporting an unstored result as successfully saved. 4. If the latest prediction is stale, the interface shall show its timestamp so users can identify the age of the result.

3.5 Flood Mapping

3.5.1 View Map with Risk Overlay

Use Case Name	View map with risk overlay
Scenario	User view the flood risk map to see all monitored locations and their latest risk levels.
Triggering Event	User opens the weather and flood risk map from the navigation menu.
Brief Description	The system displays an interactive map with all active monitored locations, showing the current flood risk level for each location using a color coded marker.
Primary Actor	General User
Secondary Actor(s)	Admins
Stakeholders	Public users, Fiji MET, Communities, Disaster response agencies
Pre-conditions	System is running and location data is available in the database.
Post-conditions	Map is displayed with monitored locations with risk levels. User may proceed to search, filter or select a location.

Use Case Name	View map with risk overlay
Scenario	User view the flood risk map to see all monitored locations and their latest risk levels.
Triggering Event	User opens the weather and flood risk map from the navigation menu.
Brief Description	The system displays an interactive map with all active monitored locations, showing the current flood risk level for each location using a color coded marker.
Primary Actor	General User
Flow of Activities	Actor: 1. The user opens the Weather and Flood Risk Map from the navigation menu. 2. The user waits for the monitored locations to load on the map. 3. The user may research for a location or filter by risk level. 4. The user may select a location marker to view more details. System: 1.1 The system displays the WeatherMapPage. 1.2 The system retrieves all active monitored location from the database. 2.1 The system displays the location on the interactive map using color coded markers based on the latest prediction level (LOW, MODERATE, HIGH, VERY_HIGH). 3.1 The system updates the displayed markers based on the users search. 4.1 The system highlights the selected location and shows a brief location card.
Exception Conditions	1. If no monitored locations are available, the system displays a message: “No monitored locations available at this time”. 2. If the map fails to load, the system displays an error message and allows the user to retry. 3. If no locations match the search or filter, the system displays a message: “No locations match your search criteria”.

3.5.2 View Location Detail

Use Case Name	View Location Detail
Scenario	User views detailed information for a selected monitored location.
Triggering Event	User selects a location marker on the map or clicks a location from the list.
Brief Description	The system displays detailed information for the selected location, including weather information, latest flood risk predictions and any active alerts or official warnings.
Primary Actor	General User
Secondary Actor(s)	Admins
Stakeholders	Public users, Fiji MET, Communities, Disaster response agencies
Pre-conditions	A monitored location is selected by the user.
Post-conditions	Location details are displayed. User may return to the map,search for another location or view related information.
Flow of Activities	Actor: 1. The user selects a location marker on the map or clicks a location from the list. 2. The user views the displayed location details. 3. The user may return to the map or select another location.

Use Case Name	View Location Detail
Scenario	User views detailed information for a selected monitored location.
Triggering Event	User selects a location marker on the map or clicks a location from the list.
Brief Description	The system displays detailed information for the selected location, including weather information, latest flood risk predictions and any active alerts or official warnings.
Primary Actor	General User
	System: 1.1 The system retrieves the selected location using its location_id. 1.2 The system retrieves the latest weather information, prediction details and active alerts. 1.3 The system displays the location name, division, weather data, flood risk level, confidence percentage and any official warnings. 3.1 The system allows the user to close the details and return to the map.
Exception Conditions	1. If the selected location does not exist or is inactive, the system displays a message: “Location not found or no longer active”. 2. If weather or prediction data is unavailable, the system displays “Data Unavailable” for the relevant information. 3. If no active alerts exist, the system indicates that no official warning is currently active.

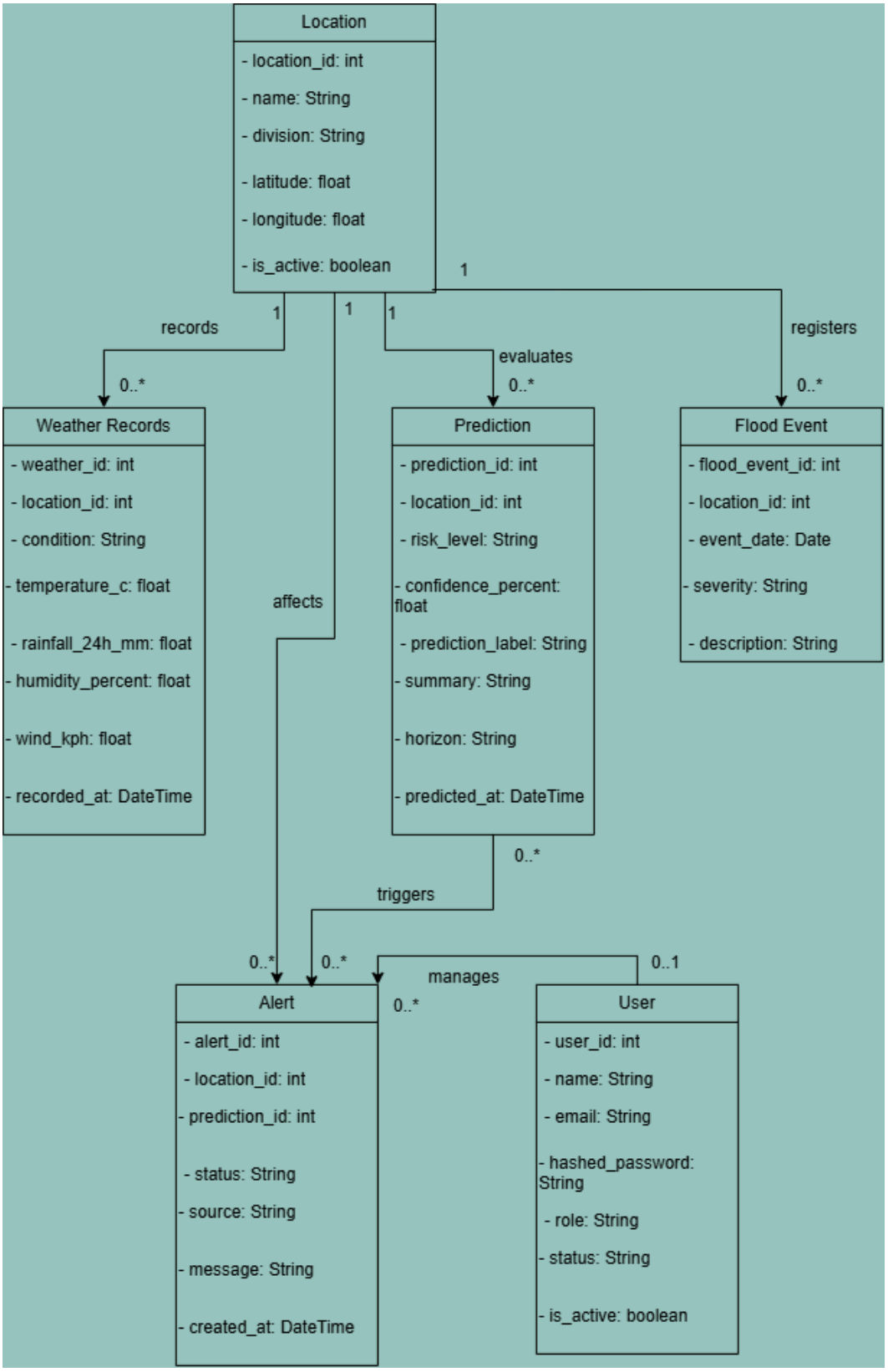
3.6 Alert System

Use Case Name	Create and Approve Flood Alert
Scenario	A High or Very High flood risk prediction is generated for a monitored location, requiring a Fiji MET admin to review and approve an alert before it is issued to the public.
Triggering Event	The AI model generates a High or Very High risk precision, or a Fiji MET administrator manually creates an alert for a monitored location.
Brief Description	The AI model generates a pending alert based on elevated flood-risk prediction or a manual admin request. The Fiji MET admin reviews the alert. Once approved, the system records the approval and makes alerts available to the public and eligible notification recipients.
Primary Actor	Fiji MET Administrator
Secondary Actor(s)	AI Prediction Service; Alert Management System; Database; Public dashboard; Notification service
Stakeholders	Fiji MET administrators, public users, at-risk communities, system administrators.
Pre-conditions	1. The Fiji MET administrator is authenticated and has the required permissions. 2. The monitored location exists and is active. 3. For an AI-generated alert, a valid flood-risk prediction is available. 4. The backend and database services are operational.
Post-conditions	1. The alert is approved and assigned an active status 2. The approving administrator and approval timestamps are recorded. 3. The approved alert is displayed on the public dashboard and flood risk map. 4. Users who have opted in to notifications for the affected location.
Flow of Activities	**ACTOR** 1. Reviews a pending alert or selects the option to manually create an alert.

	2. Reviews the affected location, risk level, confidence and prediction timestamp. 3. Edits the alert message or severity if required. 4. Approves the alert. **SYSTEM** 1.1 Detects a High or Very High flood risk prediction and creates an alert or accepts a manually created alert. 1.2 Displays the pending alert to the MET administrator. 1.3 Retrieves the relevant location and prediction timestamp. 2.1 Displays the risk level, confidence, affected area and prediction timestamp 3.1 Saves any changes made to the alert message or severity 4.1 Validates the alert information. 4.2 Changes the alert status to approved. 4.3 Records the approving administrator and timestamp. 4.4 Publishes the alert to connected public clients. 4.5 Sends push notifications to eligible users who follow the affected location.
Exception Conditions	1. If required alert information is missing, the system shall prevent approval and identify the missing fields, 2. If an invalid location is selected, the system shall reject the alert and display an appropriate error message. 3. If a duplicate pending alert already exists for the same location or prediction, the system shall display the existing alert instead of creating another one. 4. If the database or server is temporarily unavailable during approval, the system is temporarily unavailable and retry action when possible. 5. If push notification delivery fails, the system shall retry the notification and log the failure without preventing the public alert from being displayed.

4. Detailed Class Diagram (Entity Classes Only)

4.1 Class Diagram



4.2 Explanations of Entities & Relationships

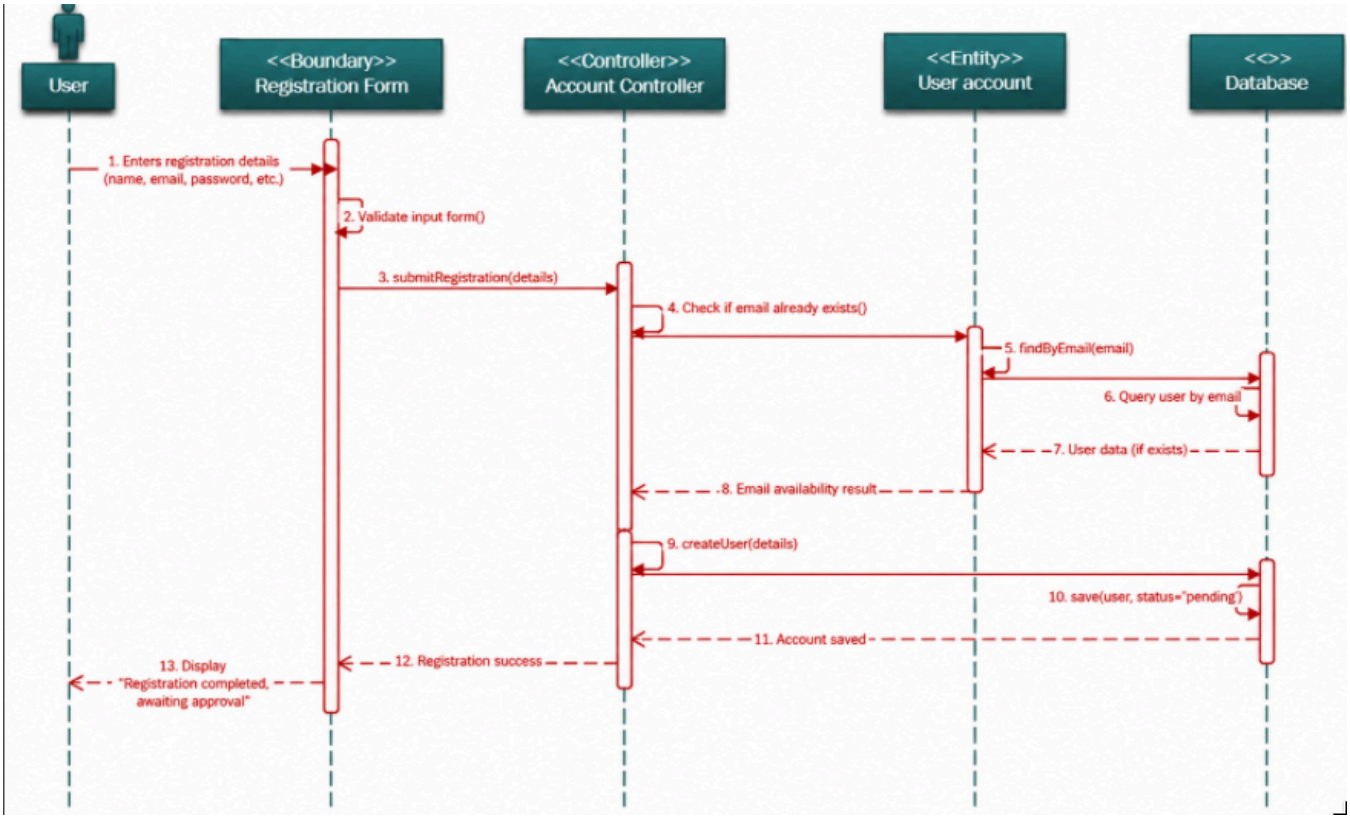
- User - it represents the people who will use the system(MET Staff & Public Users) and it stores all account credentials e.g Public/MET Staff, pending/approved and active/inactive.

- Location - are the locations of Fiji which will be monitored and on which the weather data will be based on. It has all the necessary attributes of geographical parameters like latitude, longitude and division that are used for PostGIS.
- Weather Record - this stores all weather related data like rainfall & temperature for each location. Each location can have zero to many historical weatherRecords of instances.
- Prediction - this has the predictions and outputs from the AI model. Each output has a risk level of either Low, Moderate, High or Very High with percentages. Each Location has zero to many risk predictions.
- Alert - it is responsible for the community notifications sent/shown to the users. Each location can have zero to many predictions and each prediction can have zero to many alerts. And all of these go along with their status of either Pending, Approved or Suppressed.
- Flood Event - tracks all the past history of floods and real world events across the monitored locations.

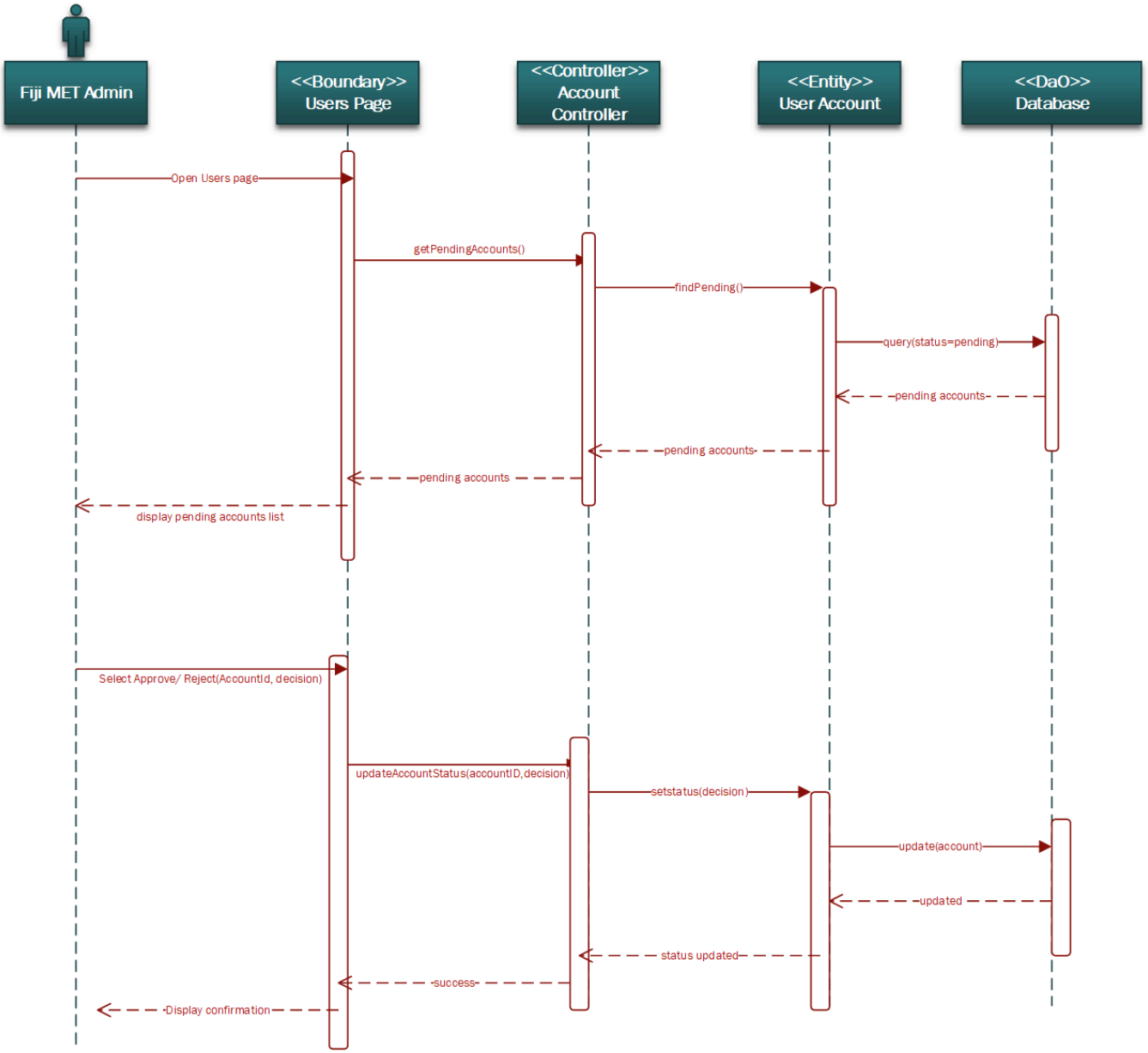
5. Sequence Diagrams

5.1 Authentication / Access (if still in approved scope)

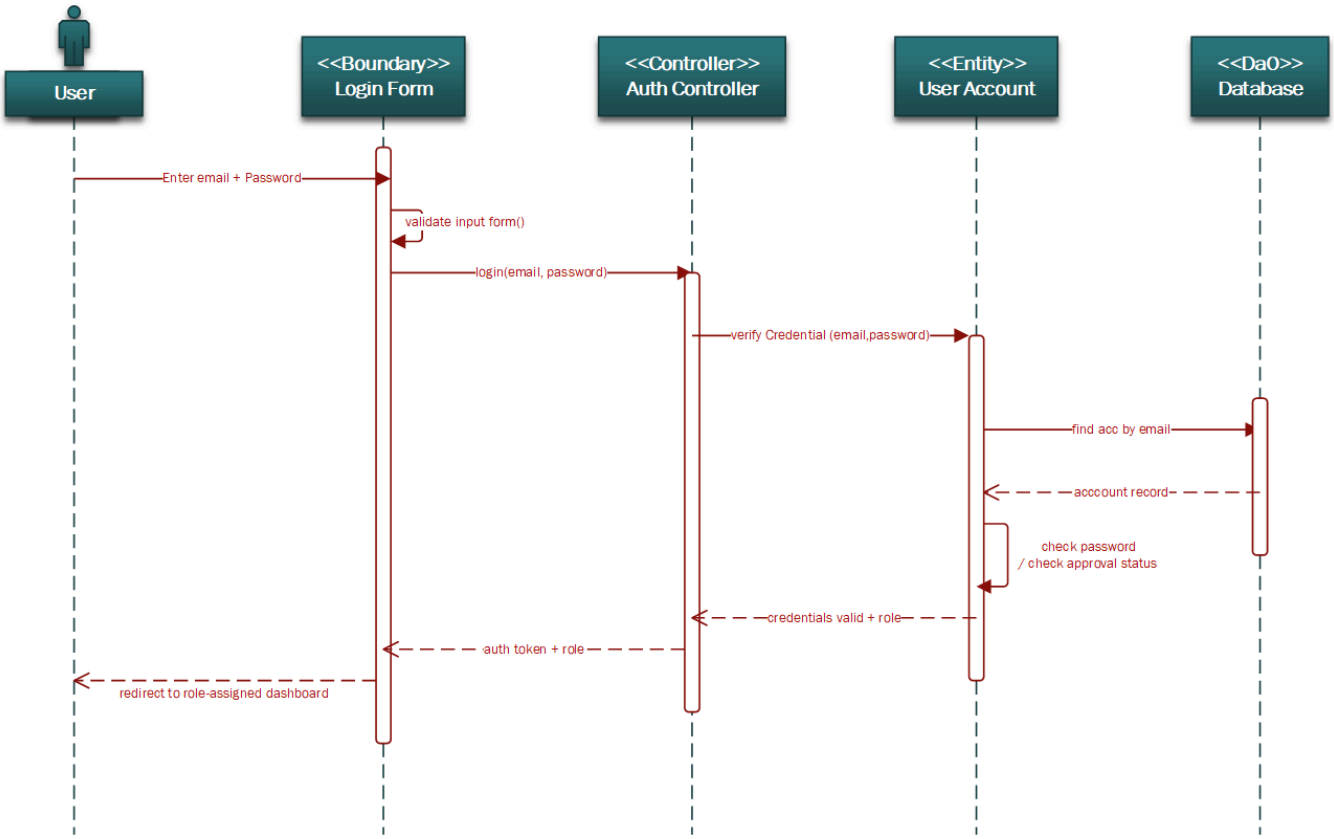
5.1.1 Registering Account



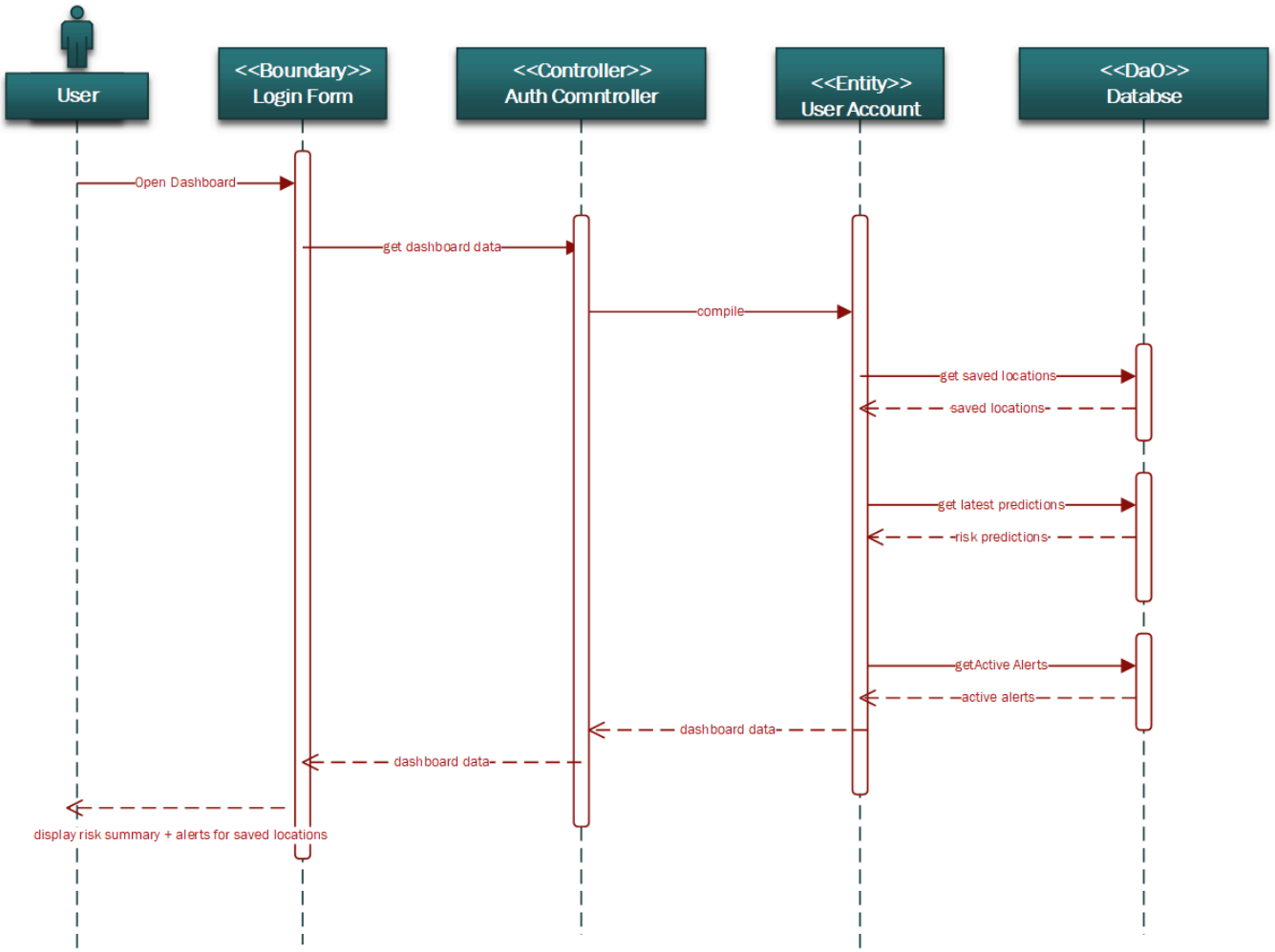
5.1.2 Approve User Account



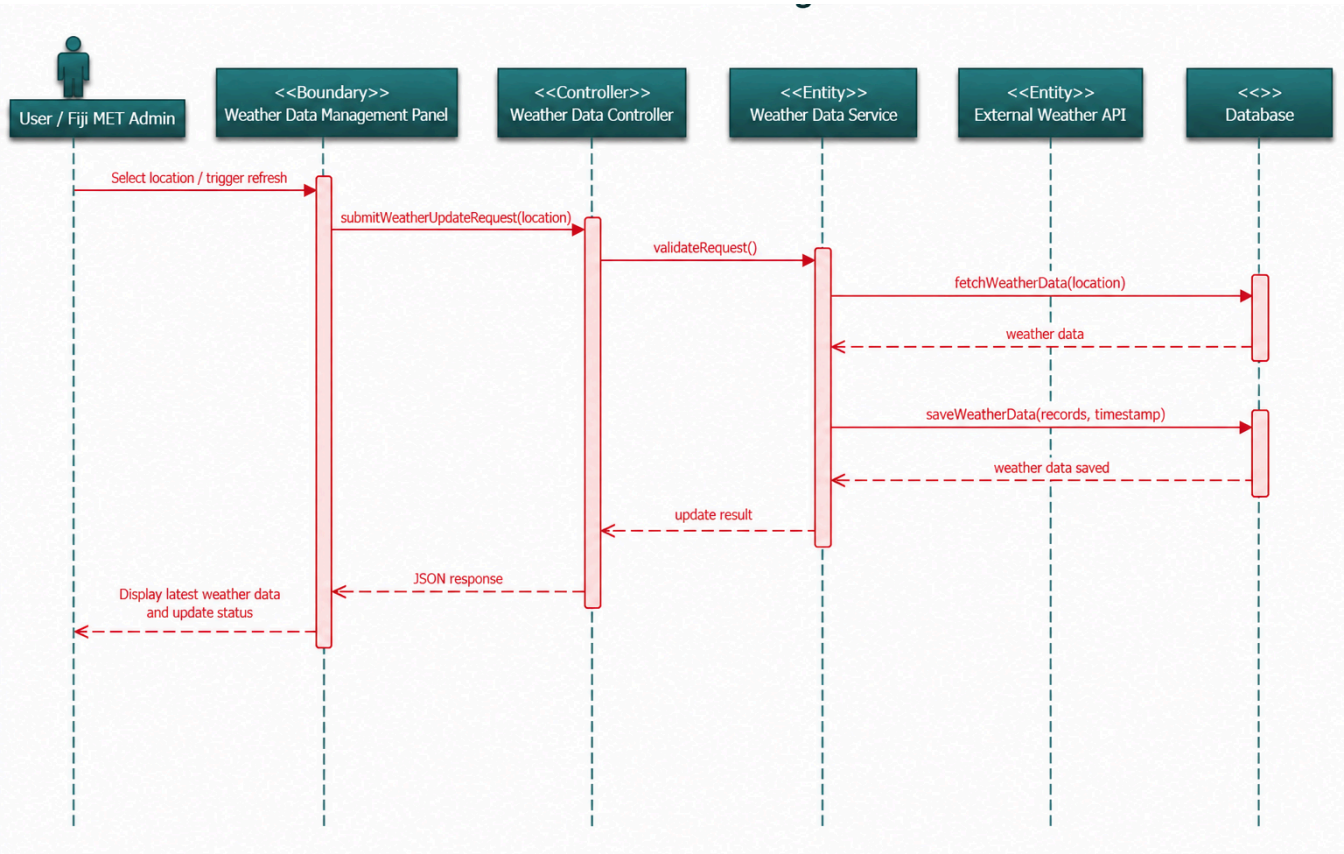
5.1.3 User Login



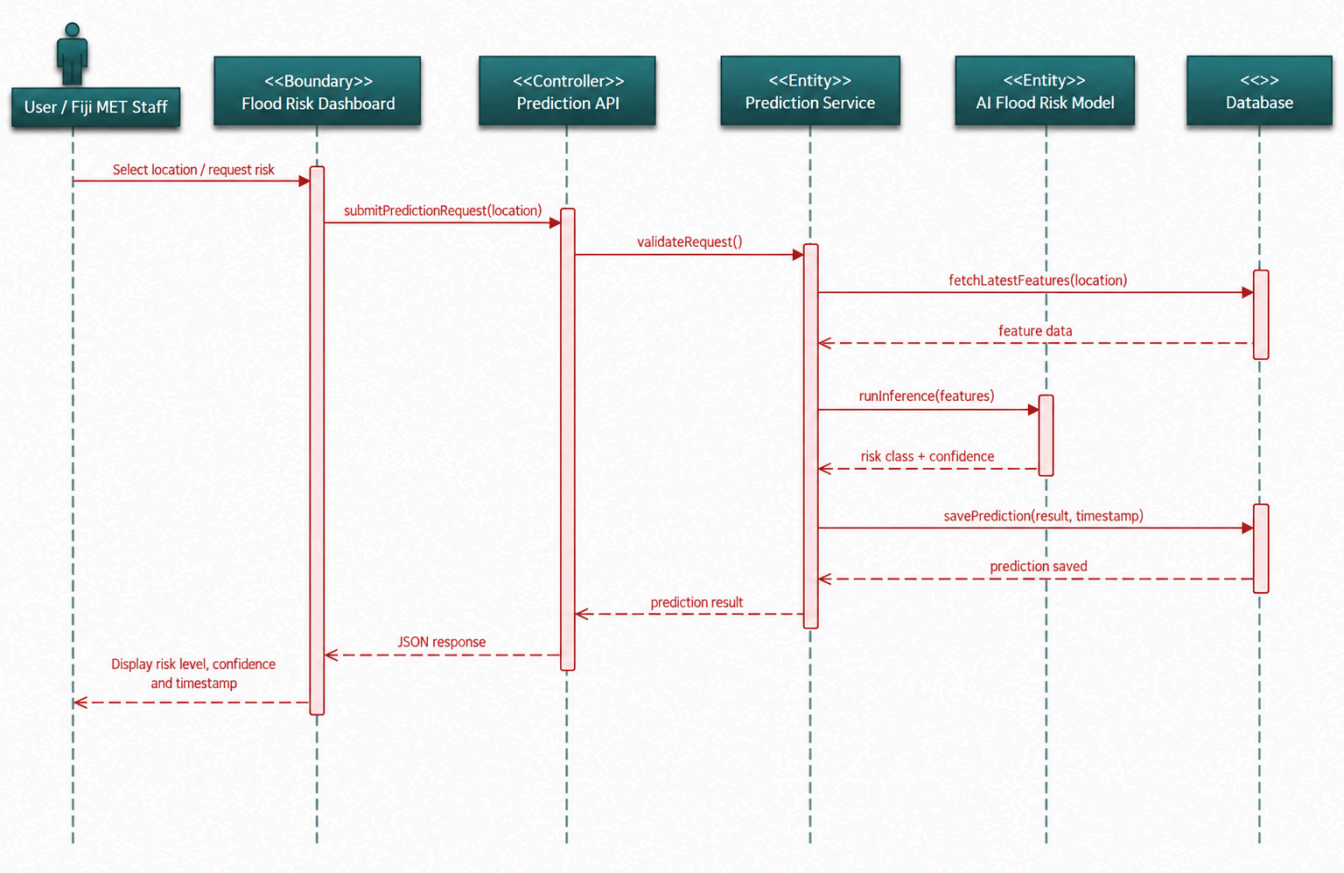
5.2 Flood Risk Dashboard



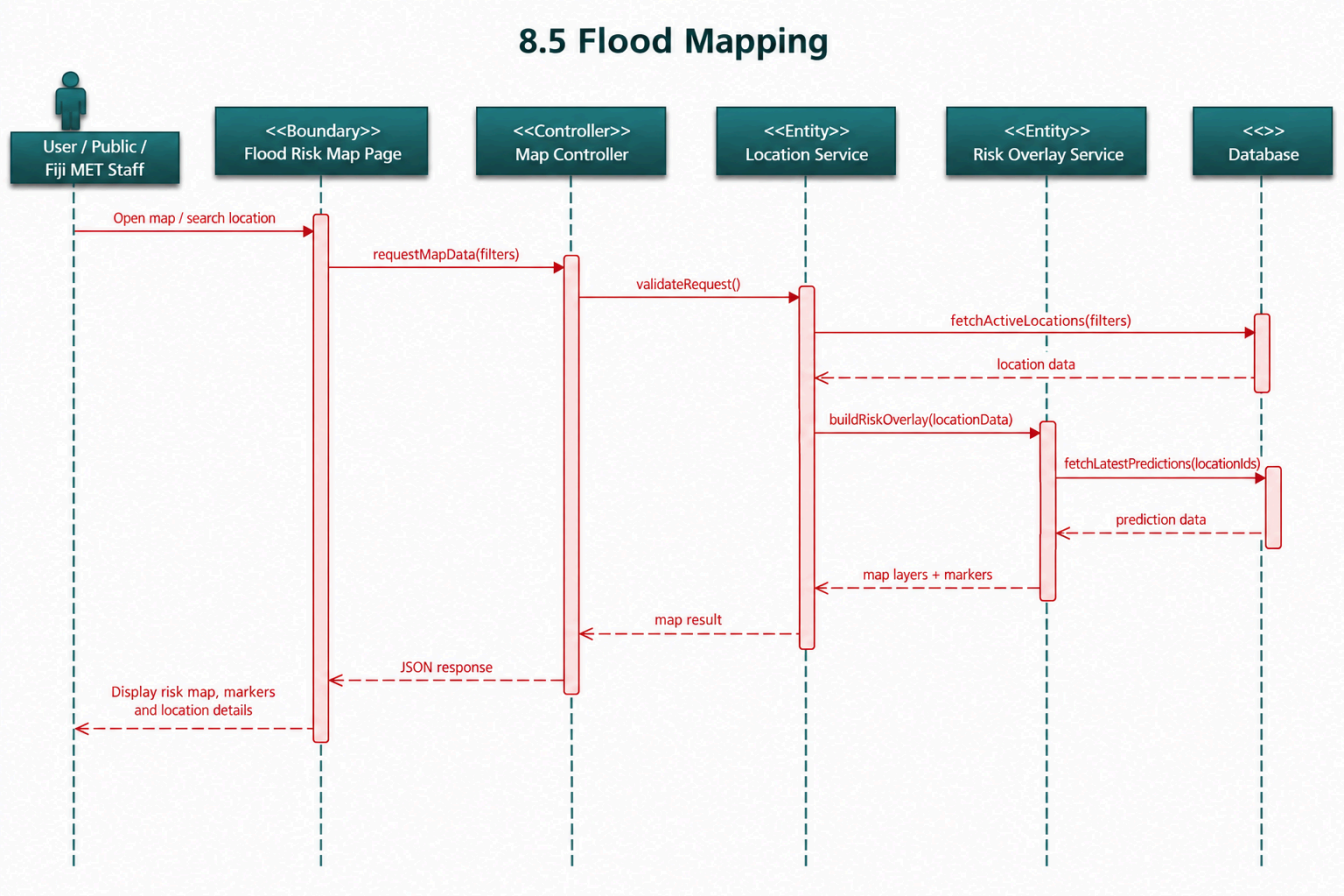
5.3 Weather Data Management


User / Fiji MET Admin
<<Boundary>>
Weather Data Management Panel
<<Controller>>
Weather Data Controller
<<Entity>>
Weather Data Service
<<Entity>>
External Weather API
<<>>
Database
Select location / trigger refresh
submitWeatherUpdateRequest(location)
validateRequest()
fetchWeatherData(location)
weather data
saveWeatherData(records, timestamp)
weather data saved
update result
JSON response
Display latest weather data
and update status

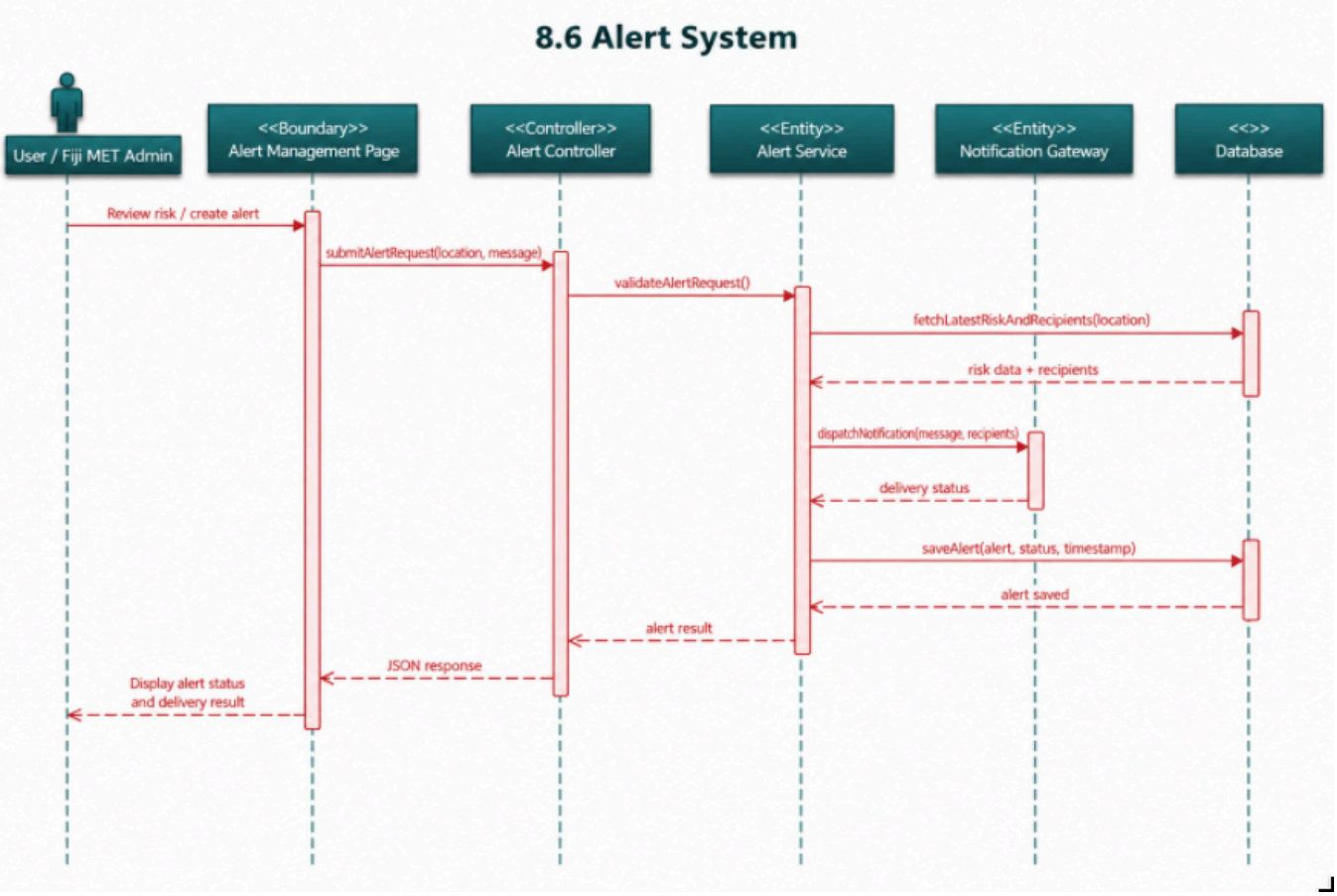
5.4 AI Flood Risk Prediction



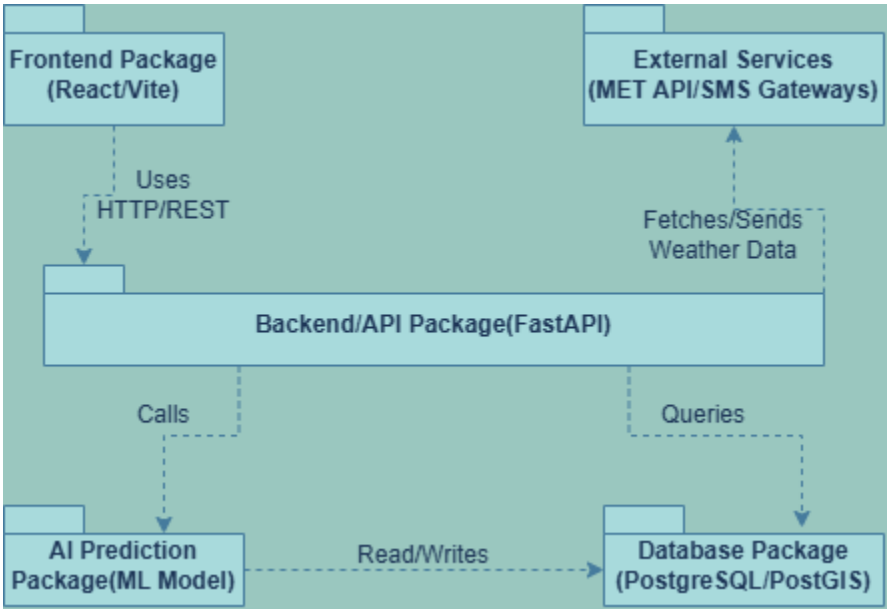
5.5 Flood Mapping


8.5 Flood Mapping
User / Public / Fiji MET Staff
<<Boundary>> Flood Risk Map Page
<<Controller>> Map Controller
<<Entity>> Location Service
<<Entity>> Risk Overlay Service
<<>> Database
Open map / search location
requestMapData(filters)
validateRequest()
fetchActiveLocations(filters)
location data
buildRiskOverlay(locationData)
fetchLatestPredictions(locationIds)
prediction data
map layers + markers
map result
JSON response
Display risk map, markers and location details

5.6 Alert System

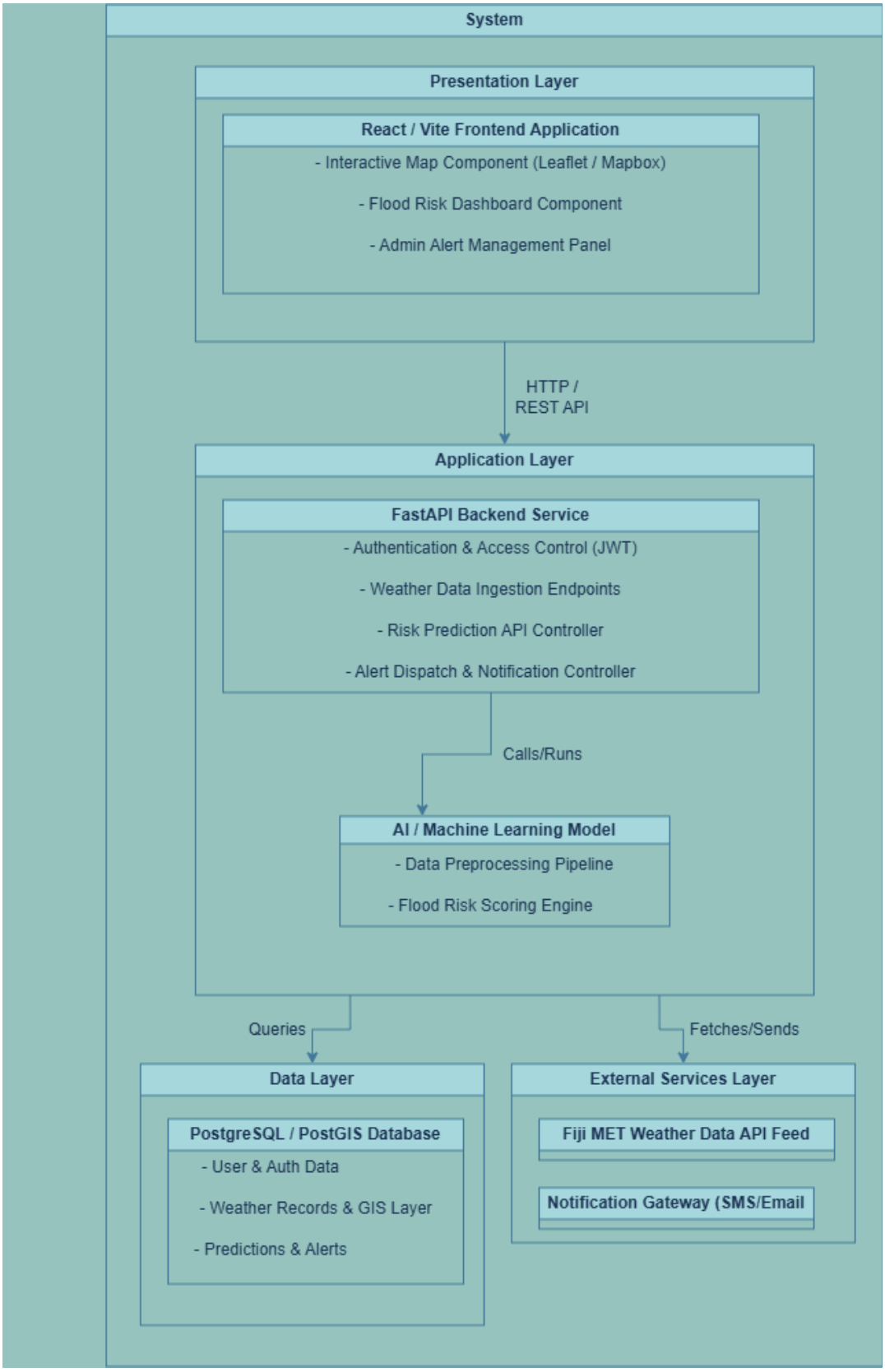


6.0 Package Diagram



The package diagram above shows how this system is put together and packaged. It shows the high level code structure and the components and its dependencies for this system. It has a React/Vite frontend which talks with the backend FastAPI through the REST API endpoints. The backend which is the FastAPI is the core of this system which acts as the main module which operates the entire system. It communicates with the External Weather API sources in fetching weather data and then Communicates that weather data after validation to the database package which is PostgreSQL and PostGIS for storing of the data. After it is validated and stored in the data base, the FastAPI calls the AI model to go look into those data set and make predictions off of it. Then the AI model reads the available current weather data set in the database and evaluates and makes predications and again stores those predictions in the Database. After that, the Backend, FastAPI takes those predictions from the database and outputs it via the frontend to the MET Staff and Public Users.

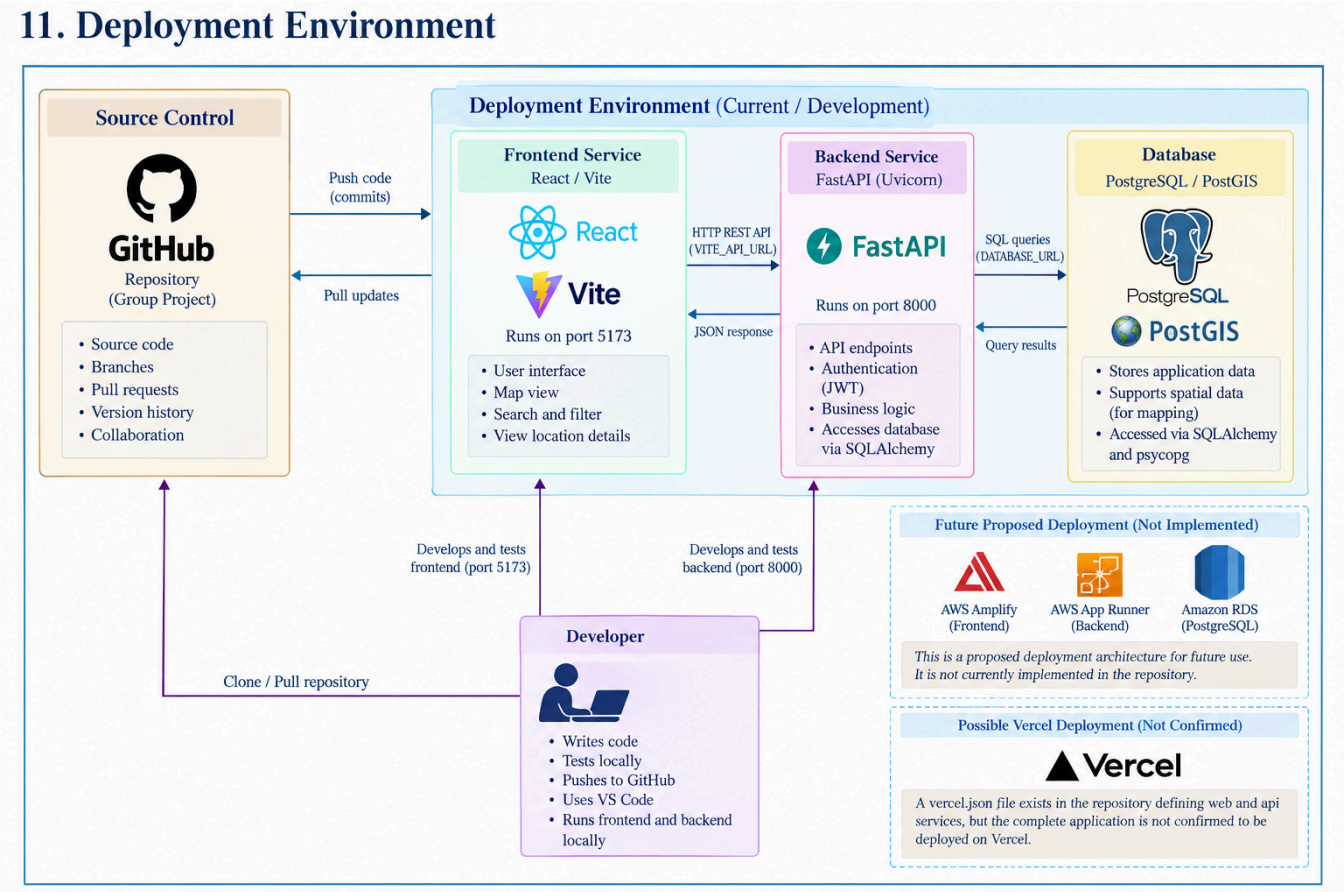
7.0 Application Architecture



This system, the Flood Risk Prediction and Community Alert System use a multi layered approach where there are multiple components in this architecture. It is divided into the 3 layers as shown above:

- **Application Layer**: this is the Backend + AI model layer which handles majority of the communication between all the components via REST API and includes the logic behind everything and the endpoints routing(everthing is connected through the Backend). The AI model is a separate model that is responsible for data reading, analyzing and making predictions based on the data provided using machine learning algorithms to make those predictions.
- **Data Layer**: this has the PostgreSQL and PostGIS databases which are responsible for storing all the weather data information and also the geographical data information alongside it which comes from the external services.
- **External Services**: these are the Fiji MET services that are provided to the system with the likes of the weather feeds that includes all the weather data and SMS/Email gateways to spread alerts and warnings.

11. Deployment Environment



The present system is based on separate Front End, Back End and Database. The Frontend is built using React and Vite and can be run locally on port 5173 and the Backend, using FastAPI and Uvicorn, runs

locally on port 8000. The frontend communicates with the backend via HTTP REST API requests and backend address is set via the environment variable VITE_API_URL.

The backend communicates with a PostgreSQL database via SQLAlchemy and psycopg driver. Database access is set up using the DATABASE_URL environment variable. Implemented backend data types in PostgreSQL include user authentication and the completed system will also include data for the primary entities in the ERD: Location, WeatherRecord, FloodEvent, Prediction and Alert. Geographic and spatial information to be used in flood-risk mapping will be supported by PostGIS.

Git is the tool employed in the project for source control, collaboration, branching and version history. Currently, the repository is missing an operational automated deployment pipeline, Docker, Docker-Compose, CI/CD workflows, Terraform, CloudFormation, etc. The presence of a vercel.json file indicates a potential deployment structure using Vercel, but it is unclear if the entire system is deployed on Vercel or whether it is part of a specific deployment. AWS deployment architecture for the project is also described in the project documentation using AWS Amplify, AWS App Runner and Amazon RDS. These are AWS services that are recommended for future implementation on production and do not exist within the implemented environment.

12. Development Environment

The Flood Risk Prediction and Community Alert System is developed using a separated frontend, backend and database environment. The frontend uses the React library with Vite, and the backend uses Python and FastAPI. When developing locally, the frontend (React/Vite) is served on port **5173** and the backend (FastAPI) is served using Uvicorn on port **8000**. The frontend/backend communication is done using HTTP REST API requests on backend API address which comes from the environment variable VITE_API_URL in the backend.

The backend includes the core application logic such as user authentication, management of weather data, flood-risk prediction requests, flood alert management and communication with the database. REST API endpoints generated by FastAPI that can be invoked on the React frontend. The AI/Machine Learning prediction component is also coded in Python, it takes the saved weather and location data and generates flood-risk classifications that are utilized by the dashboard and map and alert features. To ensure ease of development and maintenance of the different components, the application architecture contains the presentation layer (React) and the application and AI prediction layers (FastAPI).

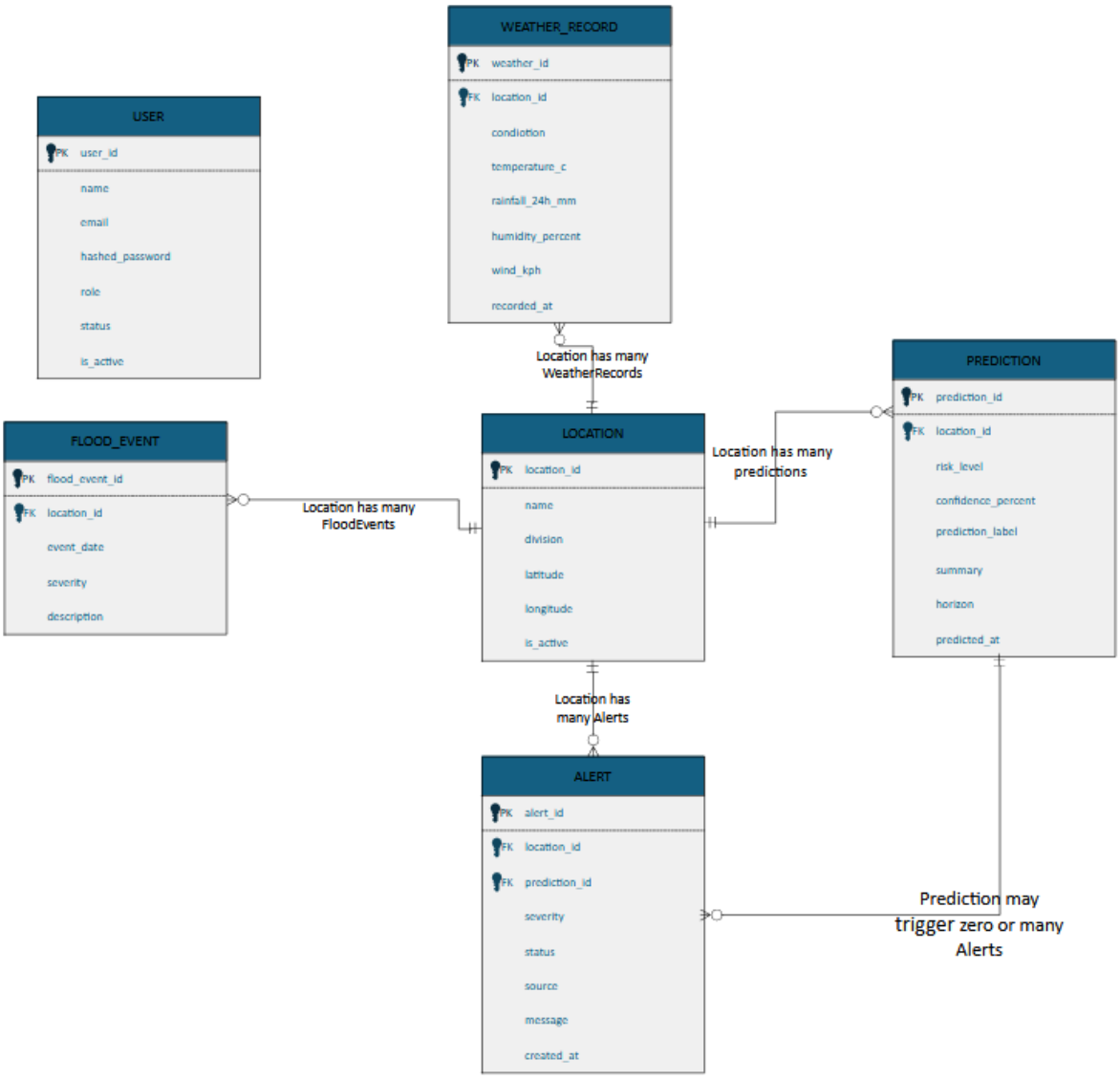
The data is stored in a PostgreSQL system, and the flood-risk mapping function includes geographical and spatial data stored in PostGIS. The FastAPI backend uses PostgreSQL via SQLAlchemy and the psycopg PostgreSQL driver. The DATABASE_URL environment variable is used to configure the database connection details, instead of hard-coding them in the source code. All system information such as users, monitored locations, weather data, historical flood data, AI predictions and alerts are stored in the database.

Source control and team collaboration are done with Git and GitHub. Branches to separate development work to allow team members to work on their piece of functionality without touching the main (stable) branch. Changes are committed and pushed to the shared GitHub repository where they can be reviewed before committing the main repository. This process keeps an upkeep of a historical record of the project, it can be possible to trace changes back to the developers and reduces the introduction of unstable code right into a primary project.

Testing and debugging is done at every step of the development process, both at the front end and backend. The team validates frontend behaviour and API communication during local development and identifies backend request, validation and application errors by FastAPI/Uvicorn server output. PostgreSQL, SQLAlchemy and backend errors provide diagnose data bases related issues. Each feature is test on its own individual development branch before merged to the main branch to avoid finding bugs in the changes when combined with the rest of the system.

The back-end API address, database connection information and other configuration values are provided by environment variables. Data repositories such as database passwords, API keys and authentication tokens thus must not be part of the source code; they must not be uploaded to the public GitHub repository. This is a development environment that enables the front end, back end, AI element and database to be developed and tested separately and to ultimately integrate via the system's REST API architecture.

10.0 Entity Relationship Diagram



11.0 Data Dictionary

Owner: Atharv | Reviewer: Rishi

Create one table like this for every persistent table/entity shown in the ERD.

11.1 User Table

#	Attribute Name	Type	Format	Required	Key	Referenced Table	Description
1	user_id	INTERGER	Whole Number	Yes	PK	-	Unique identifier for each registered user.
2	name	VARCHAR(255)	Text	Yes	-	-	Full name of the user.
3	email	VARCHAR(320)	Email address	Yes	-	-	Unique email used for authentication.
4	hashed_password	VARCHAR(255)	Encrypted/hashed text	Yes	-	-	Securely stored password hash.
5	role	VARCHAR	COMMUNITY, MET_ADMIN	Yes	-	-	Defines the users system role.
6	status	VARCHAR	PENDING, APPROVED, REJECTED	Yes	-	-	Current account approval status.
7	is_active	BOOLEAN	True/Fale	Yes	-	-	Indicates whether the user account is currently active.

11.2 Location Table

#	Attribute Name	Type	Format	Required	Key	Referenced Table	Description
1	location_id	INTEGER	Whole Number	Yes	PK	-	Unique identifier for each monitored location.
2	name	VARCHAR(100)	Text	Yes	-	-	Name of the monitored location in Fiji.
3	division	VARCHAR(100)	Text	Yes	-	-	Fiji administrative division containing the location.
4	latitude	DECIMAL	Geographic coordinate	Yes	-	-	Latitude used to position the location on the flood risk map.
5	longitude	DECIMAL	Geographic coordinate	Yes	-	-	Laongitude used to position the location on the flood risk map.
6	is_active	BOOLEAN	True/False	Yes	-	-	Determines whether the location is actively monitored and displayed.

11.3 WeatherRecord Table

#	Attribute Name	Type	Format	Required	Key	Referenced Table	Description
1	weather_id	INTEGER	Whole Number	Yes	PK	-	Unique identifier for each weather observation.
2	location_id	INTEGER	Whole Number	Yes	FK	Location	Identifies the monitored location associated with Weather Record.
3	condition	VARCHAR(100)	Text	No	-	-	Current weather condition such as Heavy Showers.
4	temperature_c	DECIMAL	Degrees Celsius	No	-	-	Recorded air temperature in Celsius.
5	rainfall_24h_mm	DECIMAL	Milimetres	Yes	-	-	Rainfall accumulated during the previous 24 hours.
6	humidity_percent	DECIMAL	Percentage	No	-	-	Recorded relative humidity percentage.
7	wind_kph	DECIMAL	Kilometres per hour	No	-	-	Record wind speed.
8	recorded_at	TIMESTAMP	Date and Time	Yes	-	-	Date and time when the weather observation was record.

11.4 FloodEvent Table

#	Attribute Name	Type	Format	Required	Key	Referenced Table	Description
1	flood_event_id	INTEGER	Whole Number	Yes	PK	-	Unique identifier for a historical or recorded flood event.
2	location_id	INTEGER	Whole Number	Yes	FK	Location	Identifies where the flood event occurred.
3	event_date	TIMESTAMP	Date and Time	Yes		-	Date and time associated with the flood event.
4	severity	VARCHAR(20)	Risk category	No		-	Severity of the recorded flood event.
5	description	TEXT	Free text	No		-	Additional information describing the flood event.

11.5 Prediction Table

#	Attribute Name	Type	Format	Required	Key	Referenced Table	Description
1	prediction_id	INTEGER	Whole Number	Yes	PK	-	Unique identifier for each AI-generated prediction.
2	location_id	INTEGER	Whole Number	Yes	FK	Location	Identifies the location for which the prediction was generated.
3	risk_level	VARCHAR(20)	LOW, MODERATE, HIGH, VERY_HIGH	Yes	-	-	Flood risk category generated by the AI-model.
4	confidence_per cent	DECIMAL	0-100 percentage	Yes	-	-	Confidence level produced by the AI-prediction model.
5	prediction_label	VARCHAR(25 5)	Text	No	-	-	Human readable description of the prediction such as High River Flood Potential.
6	summary	TEXT	Free text	No	-	-	Explanation or summary of the predicted flood conditions.
7	horizon	VARCHAR(50)	Text/time period	No	-	-	Forecast period represented by the prediction, such as Next hours.
8	predicted_at	TIMESTAMP	Date and time	Yes	-	-	Date and time when the prediction was generated.

11.6 Alert Table

#	Attribute Name	Type	Format	Required	Key	Referenced Table	Description
1	alert_id	INTEGER	Whole Number	Yes	PK	-	Unique identifier for each generated warning or alert.
2	loaction_id	INTEGER	Whole Number	Yes	FK	Location	Identifies location affected by the alert.
3	prediction_id	INTEGER	Whole Number	No	FK	Prediction	Identifies the AI prediction associated with the alert where applicable.
4	severity	VARCHAR(20)	LOW, MODERATE, HIGH, VERY_HIGH	Yes	-	-	Severity of the alert.
5	status	VARCHAR(20)	PENDING, APPROVED, REJECTED, SUPPRESSED	Yes	-	-	Current state of the alert.
6	source	VARCHAR(30)	AI_RECOMM ENDATION,	Yes	-	-	Indicates whether the

#	Attribute Name	Type	Format	Required	Key	Referenced Table	Description
			MET_OFFICIAL				alert originated from the AI system or Fiji MET.
7	message	TEXT	Free text	Yes	-	-	Warning message displayed to users.
8	created_at	TIMESTAMP	Date and time	Yes	-	-	Date and time when the alert was created.

12.0 Functional Requirements

12.1 Authentication / Access (if in scope)

REQ-1: The system shall require that users need to have a valid, unused email and a strong password to register

REQ-2: The system shall prevent the creation of multiple accounts using the same email.

REQ-3: The system shall store all user data in the secure PostgreSQL database.

REQ-4: The system shall allow for Fiji MET admin to view active users and pending account requests.

REQ-5: The system shall update account request status (approve/reject) based on admins professional decision.

REQ-6: The system shall deny specific system functions until an account is approved.

REQ-7: The system shall verify user credentials against stored credentials.

REQ-8: The system shall issue authentication token upon users successful login

REQ-9: The system shall redirect users to their role-based dashboard upon successful login.

12.2 Flood Risk Dashboard

REQ-10: The system shall display personalized risk summary based on each location saved by the user.

REQ-11: The system shall display active alerts pertaining to users saved location.

REQ-12:The system shall allow users the ability to add a location to a saved list.

REQ-13:The system shall allow a user to remove a location from their saved lists.

12.3 Weather Data Management

REQ-14:The system shall automatically download weather data for all monitored locations within a predefined interval.

REQ-15:The system shall have the ability to retrieve and use external weather data API/source.

REQ-16:The system shall always be stamped on every retrieval attempt.

REQ-17:The system shall allow an admin to manually trigger data retrieval for specific locations

REQ-18:The system shall validate all required data fields before storing records.

REQ-19:The system shall ensure that all weather records are within a plausible range (e.g non-negative values for temperature).

REQ-20:The system shall reject and log data that fails validation without storing it for consistency.

REQ-21:The system shall prevent having any duplicate weather record for different locations and timestamps.

12.4 AI Flood Risk Prediction

REQ-22:The system shall process past and present meteorological data provided by Fiji MET

REQ-23:The system shall generate flood risk predictions from processed data.

REQ-24:The system shall organize and categorize flood-risk predictions by urgency level (Low/Moderate/High/VeryHigh).

REQ-25:The system shall update predications in a predetermined time range, or when new processed meteorological data becomes available.

REQ-26:The system shall allow users to specify locations via space bar or by interacting with a live interactive map,

REQ-27:The system shall display flood-risk predictions levels for the selected locations.

REQ-28:The system shall display the flood-risk levels for the users selected location..

12.5 Flood Mapping

REQ-29:The system shall display flood-risk levels with the interactive map, associating different colors to urgency level.

REQ-30:The system shall update the maps overlap in real time when new predictions are made.

REQ-31:The system shall have a fully interactive map allowing users to zoom and pan to specific locations.

REQ-32:The system shall allow for users to select specific locations to view its flood-risk predictions.

REQ-33:The system shall display current flood-risk levels and other past meteorological data for the specified selected location.

12.6 Alert System

REQ-34: The system shall automatically create a pending alert when the risk is “High” and “Very High” for a monitored location.

REQ-36: The system shall allow for admins to manually create alerts that are independent of a prediction.

REQ-37: The system shall display all pending alerts with the necessary information like risk level,location and timestamp.

REQ-38: The system shall allow for Admins to edit risk alerts before approval.

REQ-39: The system shall record all logs regarding approval and denial of risk alerts.

REQ-40: The system shall allow for Admins to suppress alerts from public view, also attach reason for documentation purpose.

REQ-41: The system shall update all users on any overridden or suppressed notifications to keep users informed.

REQ-42: The system shall display all active, approved alerts to all users, including non registered users.

REQ-43: The system shall use color coding to distinguish risk level severity.

REQ-44: The system shall allow for users to opt in and out to receive push notification for saved locations.

REQ-45: The system shall send notifications to opted users minutes after alerts are approved and posted.

REQ-46: The system shall not send suppressed alerts to users

12.7 Administrative / Data Management requirements (if in scope)

REQ-47: The system shall allow MET admin to edit monitored location data and specification.

REQ-48: The system shall only present validated location information before Admin edit.

REQ-49: The system shall require admin confirmation before removing monitored location

REQ-50: The system shall restrict location management to Fiji MET admin users

REQ-51: The system shall only display system metrics to Fiji MET admin users in the Admin Dashboard.

REQ-51: The system shall restrict analytics access to authorized Fiji MET admin users

REQ-51: The system shall allow report generation for Fiji MET admin user

13.0 Non-functional Requirements

13.1 Performance

NREQ-1: The system shall return dashboard and map data within 3 seconds under normal load and within 5 seconds under peak load (e.g. during an active severe weather event).

NREQ-2: The system shall generate a flood risk prediction for a given location within 5 seconds of a request or scheduled trigger.

NREQ-3: The system shall propagate a newly approved, overridden or suppressed alert to all connected clients.

NREQ-4: The system shall support at least 150 concurrent users browsing the public dashboard and flood-risk map without noticeable performance degradation.

NREQ-5: The system shall complete scheduled weather-data ingestion and validation for all monitored locations within the configured polling interval (e.g. hourly) without blocking other system operations such as admin displays.

13.2 Security and Privacy

NREQ-6: The system shall require users to authenticate before accessing any account-specific or administrative feature (e..g alert approval, threshold configuration, monitored location management).

NREQ-7: The system shall enforce role-based access control restricting MET Admin functions (user approval, alert approval/override suppression, threshold configuration) to accounts with the ADMIN role.

NREQ-8:The system shall store all user passwords as salted cryptographic hashes and never store or log plaintext passwords.

NREQ-9: The system shall issue time-limited authentication tokens (e.g. JWT) and reject expired or invalid tokens on every protected request.

NREQ-10:The system shall not transmit all data between client and server over HTTPS or TLS to users.

NREQ-11: The system shall not expose internal error details, stack traces, or database errors to end users.

NREQ-12: The system shall log all administrative actions on user accounts and alerts (who performed the action and when) to support accountability and audit review.

NREQ-13:The system shall handle user and location data in accordance with Fiji's required data protection acts (e.g. CERT guidance) and collect only the data necessary for system operation.

13.3 Usability

NREQ-14: The system shall present flood risk information (risk-levels, alerts, warnings) in plain, non-technical language with clear colour-coded indicators, understandable to users without a meteorological background,

NREQ-15: The system shall use a responsive layout that remains fully usable on desktop and mobile screen sizes.

NREQ-16: The system shall maintain consistent navigation, layout and terminology across all public and administrative pages

NREQ-17: The system shall allow a user to reach any key public feature (dashboard, map, alerts) in no more than 3 clicks from the landing page.

NREQ-18: The system shall clearly distinguish AI-generated predictions from official Fiji MET/government warnings on every screen where both appear,

13.4 Reliability / Availability

NREQ-19: The system shall be available at least 99% of the time during normal operating conditions, excluding scheduled maintenance.

NREQ-20: If an external weather data source is unavailable, the system shall display the most recently successfully retrieved data along with a timestamp, rather than showing no or stale data as current.

NREQ-21: The system shall recover from critical failure (application or database outage) within 1 hour without loss or corruption of stored weather, prediction, or alert data.

NREQ-22: The system shall retry failed weather data retrieval attempts according to a defined retry policy and log each attempt's outcome.

13.5 Maintainability / Scalability

NREQ-23: The system shall maintain a clear separation between the frontend (React), backend/API (FastAPI), and AI prediction component, so that each can be updated or redeployed to the application codebase.

NREQ-24: The system shall allow new monitored locations to be added without requiring changes to the application codebase.

NREQ-25: The system shall be deployable via PostgreSQL and FastAPI to support scaling of backend and database resources as the number of users and monitored locations increase.

NREQ-26:The system shall be deployable via consistent coding conventions including documents sufficient for a new developer to understand and extend a module (e.g. adding a new alert channel) without major rework of existing components.

13.6 Safety / Data Integrity

NREQ-27: The system shall prevent any account other than ADMIN roles from approving, overriding, or suppressing a public alert.

NREQ-28: The system shall prevent historical data records and past flood-risk predictions from being modified or deleted by any user, including ADMIN roles.

NREQ-29: The system shall validate all incoming weather data against expected ranges (e.g. non-negative rainfall) before storing it, and shall reject and log any record that fails validation.

NREQ-30: The system shall prevent duplicate pending alerts from being created for the same location and prediction.

NREQ-31: The system shall require an ADMIN account to remain in PENDING status until explicitly approved or rejected by another authorized admin.

14.0 Conclusion

The document detailing Systems Analysis and Design for the AI-Powered Flood Risk Prediction and Community Alert System for Fiji MET provides a comprehensive outline of the proposed system's architecture and how its key functionalities will interact with one another. It includes information on the system's use cases, class design, operational sequences, package structure, overall application architecture, environments for deployment and development, database design, as well as both functional and non-functional specifications.

The proposed system comprises several key elements, such as the Flood Risk Dashboard, Weather Data Management, AI-driven Flood Risk Prediction, Flood Mapping, the Alert System and Access Management. These components collaborate to gather and manage meteorological data, produce flood risk forecasts, present risk information based on geographic locations, and relay pertinent alerts to both users and personnel at Fiji MET. The design outlines the interactions between the React/Vite frontend, FastAPI backend, AI prediction component and PostgreSQL database within the overall application architecture. The diagrams and requirements included in this document offer the development team a solid basis for implementing, integrating and testing the system. Additionally, they give stakeholders a clearer insight into the anticipated behavior and structure of the proposed solution. In summary, the

Software Architecture Document (SAD) serves as a crucial reference for the remaining stages of development and helps ensure that the final system aligns with the project's approved goals and requirements.

15.0 Appendix

15.1 Mark Allocation Sheet

After our group discussion, we recommend the following mark allocation to each group member, based on level of contribution throughout this phase of the project.

Group Number: 13

Project Manager: Parvish Mohan

Student ID	Contribution %
S11230414	100
S11225654	100
S11229870	100
S11220998	100
S11231286	100

Group Member	Student ID	Signature	Date
Parvish Mohan	S11230414	PRmohan	2026
Aryan Sewak	S11225654	Aryan Sewak	2026
Rishwant Kumar	S11229870	R.k	2026
Louvatu Vosailagi	S11220998	LVosailagi	2026

Atharv Narayan	S11231286		2026

15.2 AI Use Declaration / Log

Section / Task	AI Tool	How AI Was Used	Human Verification / Rewrite
[e.g. Introduction]	[tool]	[summary / expansion / wording support]	[who checked and what was changed]
Functional Requirements	Claude	To collate all requirements from previous documents and discussions. This was to ensure all discussed requirements are consistent across all modules. No requirements were created with AI.	Every member reviewed the requirements for their specific module.Duplicates and inconsistencies were removed.
Conclusion	Chatgpt	To assist with wordings and concluding supporting documents together. For structure purposes.	Checked by PM- Parvish